

```
import warnings
warnings.filterwarnings('ignore')
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
import matplotlib.pyplot as plt
from wordcloud import WordCloud

import nltk
nltk.download('punkt')
nltk.download('stopwords')
from nltk.corpus import stopwords
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Unzipping corpora/stopwords.zip.
```

```
data = pd.read_csv('AmazonReview.csv')
data.head()
```

	Review	Sentiment
0	Fast shipping but this product is very cheaply...	1
1	This case takes so long to ship and it's not e...	1
2	Good for not droids. Not good for iPhones. You...	1
3	The cable was not compatible between my macboo...	1
4	The case is nice but did not have a glow light...	1

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Review      24999 non-null  object
1    Sentiment    25000 non-null  int64
dtypes: int64(1), object(1)
memory usage: 390.8+ KB
```

```
data.dropna(inplace=True)
```

```
#1,2,3->negative(i.e 0)
data.loc[data['Sentiment']<=3,'Sentiment'] = 0
```

```
#4,5->positive(i.e 1)
data.loc[data['Sentiment']>3,'Sentiment'] = 1
```

```
stp_words=stopwords.words('english')
def clean_review(review):
    cleanreview=" ".join(word for word in review.
                          split() if word not in stp_words)
    return cleanreview

data['Review']=data['Review'].apply(clean_review)
```

```
data.head()
```

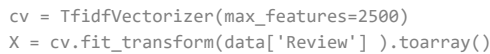
```
data['Sentiment'].value_counts()
```

```
consolidated = ' '.join(word for word in data['Review'][data['Sentiment']==0].astype(str))
wordCloud = WordCloud(width=1600,height=800,random_state=21,max_font_size=110)
plt.figure(figsize=(15,10))
plt.imshow(wordCloud.generate(consolidated),interpolation='bilinear')
plt.axis('off')
plt.show()
```



```
consolidated=' '.join(word for word in data['Review'][data['Sentiment']==1].astype(str))
wordCloud=WordCloud(width=1600,height=800,random_state=21,max_font_size=110)
plt.figure(figsize=(15,10))
plt.imshow(wordCloud.generate(consolidated),interpolation='bilinear')
plt.axis('off')
plt.show()
```





```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
# Model accuracy
accuracy = accuracy_score(y_test, pred)
print("Accuracy:", accuracy)
```

```
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
from sklearn.linear_model import LogisticRegression
```

```
# Display confusion matrix
cm_display = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["False", "True"])
cm_display.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix")
plt.show()
```

